

vasic-digital tmux — QWEN.md

Qwen Code CLI agent view of this project's governance. Base agent rules live at constitution/QWEN.md and the constitution/Constitution.md it references. **READ THOSE FIRST.** This file extends them with project-specific rules; it never weakens them. Canonical project authority: Constitution.md — Project Articles §101–§109. For the full agent rule set see AGENTS.md, which this file mirrors.

Fresh-conversation workflow: read CONTINUATION.md first (§0/§3/§8 mandatory), then Issues.md for OPEN/PARTIAL/BLOCKED/RUNNING.

MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

Operative rule. The bar for shipping is **not** "tests pass" but "**end users can actually use the feature.**" Every PASS — test OR HelixQA Challenge — **MUST** carry positive captured runtime evidence that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-without-runtime PASS are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound **EQUALLY**.

This covenant is restated verbatim in every governance file at the consumer layer (CLAUDE.md, AGENTS.md, QWEN.md) so any tool that does not expand @imports still reads it. The same verbatim block lives upstream at constitution/Constitution.md §11.4, constitution/CLAUDE.md, and constitution/AGENTS.md. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: constitution/Constitution.md §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: Constitution.md §101.

Non-compliance is a release blocker regardless of context.

Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the tmx wrapper. Native dual-OS: each `tmx new -s NAME` spawns its own tmux server with OS-native isolation — `cgroup-v2` scope on Linux, POSIX `rlimit` wrapper on macOS. Session shell is the operator's host shell with full `PATH`. No VM.

Critical base rules restated (for agents that don't follow `@imports`)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (file *content*, not existence). Universal §11.4 / Project §101.
- **FAIL-bluffs forbidden.** Fix script-bug failures at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing language.** `likely`, `probably`, `maybe`, `seems`, `appears` are forbidden when reporting causes — prove with forensic evidence or mark `UNCONFIRMED:` / `PENDING_FORENSICS:`. Universal §11.4.6.
- **Operator-path coverage.** Gate tests use `tmx new -s X`. Project §102.
- **Four-layer coverage.** Every defect lands pre-build gate + runtime test + HelixQA Challenge + paired mutation. Project §103.
- **Use the commit wrapper.** `bash commit_all.sh "<msg>"` only — no direct `git push` on the main repo.
- **Never force-push** without explicit per-operation authorization.
- **CONTINUATION.md** kept in sync in every non-trivial commit. Universal §12.10.
- **60 % RAM cap** enforced per-session via `TMX_MEM`. Project §106.

Commands (exact)

Step	Command
Install build deps	<code>bash scripts/install_deps.sh</code>
Full pipeline	<code>bash scripts/setup.sh</code>
Verification gate	<code>bash scripts/verify.sh</code>
Run all tests	<code>bash scripts/tests/run_all.sh</code>
Inheritance gate	<code>bash scripts/tests/ test_constitution_inheritance.sh</code>
Meta-test (paired mutation)	<code>bash scripts/tests/ meta_test_false_positive_proof.sh</code>
End-to-end automation	<code>bash scripts/test_e2e.sh</code>
Commit + push (github+gitlab)	<code>bash commit_all.sh "message"</code>
Per-session wrapper	<code>tmx {new attach ls kill}</code>

Project-specific agent rules

Forbidden in this project

- Editing `tmux/`, `constitution/`, or generated files (`scripts/tmx`, `tmux/build*`).
- Direct `git push` on the main repo — use `commit_all.sh`.
- CI/CD pipeline files; HTTPS Git remotes (SSH only).
- Advancing the `tmux/` submodule off tag 3.6a without an explicit documented decision (Project §107).

Project-specific workflow

- **Test-interrupt-on-discovery** (§103): a defect found mid-cycle STOPS the cycle — fix root cause, land four layers, rebuild, repeat.
- **Topology dispatch** (§104): tests detect host topology and SKIP-with-reason; never silently degrade.
- **Destructive tests** (12/13/14): require `TMX_TEST_DESTRUCTIVE=1`.

Project overrides of universal rules

None.

§11.4.87–98 — newly-inherited anchors (constitution 6828ff2)

Twelve new universal anchors landed in `constitution/Constitution.md` at commit 6828ff2 (2026-05-28). They apply unconditionally per §11.4.35.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch

Inherited from `constitution/Constitution.md` §11.4.87. Continue work until `Issues.md` has zero non-terminal items, `CONTINUATION.md` §3 empty, no in-flight subagent, no external dependency in-flight. Every closure carries §11.4.4(b) four-layer coverage + captured-evidence per §11.4.5/§11.4.69. **Canonical authority:** §11.4.87.

§11.4.88 — Background-push: commit-flock immediate release, push detached

Inherited from `constitution/Constitution.md` §11.4.88. `commit_all.sh` releases its commit flock the moment `git commit` returns 0; push runs detached. Per-remote flock for concurrency control. Only escape: `--sync-push` for §11.4.41 force-push. **Canonical authority:** §11.4.88.

§11.4.89 — Background test execution

Inherited from constitution/Constitution.md §11.4.89. Any test > 30 s MUST be detached via `nohub ... > <log> 2>&1 & + disown`; main stream returns to §11.4.42 priority queue. **Canonical authority:** §11.4.89.

§11.4.90 — Obsolete status + per-item obsolescence audit

Inherited from constitution/Constitution.md §11.4.90. New Status terminal value `Obsolete` (→ `Fixed.md`) with `**Obsolete-Details:**` audit-line (Since + Reason + Superseding-item + Triple-check evidence). **Canonical authority:** §11.4.90.

§11.4.91 — Summary-doc clarity

Inherited from constitution/Constitution.md §11.4.91. Every summary one-liner ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden anti-patterns: section labels, bare metadata, §-letter alone. **Canonical authority:** §11.4.91.

§11.4.92 — Multi-pass change-evaluation discipline

Inherited from constitution/Constitution.md §11.4.92. 5 passes before commit: main-task evidence; regression blast radius; cross-feature interaction; deep research + CodeGraph; anti-bluff confirmation. Trivial exemption only for zero-source-touch commits. **Canonical authority:** §11.4.92.

§11.4.93 — SQLite-backed single source of truth for workable items

Inherited from constitution/Constitution.md §11.4.93. Authoritative DB at `docs/workable_items.db`; Go binary `cmd/workable-items/` bidirectional sync. Round-trip byte-identical within whitespace tolerance. `commit_all.sh` refuses non-empty diff. **Canonical authority:** §11.4.93.

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode

Inherited from constitution/Constitution.md §11.4.94. Idle only when genuinely externally blocked OR operator STOP OR §12 host-safety. Before wake/sleep, survey parallel-work feasibility and dispatch non-contending items via subagent-driven default. **Canonical authority:** §11.4.94.

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored

Inherited from constitution/Constitution.md §11.4.95. Amends §11.4.93: the DB is authoritative source data, TRACKED, never gitignored. Every mutation stages + commits + pushes alongside the MD regen. WAL-checkpoint before commit-stage. **Canonical authority:** §11.4.95.

§11.4.96 — Safe-parallel-work-with-long-build catalogue

Inherited from constitution/Constitution.md §11.4.96. SAFE during long builds: docs/scripts/tests/submodule pushes, research, DB ops, subagent dispatch. UNSAFE: destructive git ops, mass deletes, host-safety breaches. Conductor consults catalogue at every pause. **Canonical authority:** §11.4.96.

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence

Inherited from constitution/Constitution.md §11.4.97. Idle minute during which work could progress and is not genuinely blocked is a §11.4.97 violation. 1-line operator-facing update at every commit / subagent return / anchor landing / evidence acquisition / milestone. **Canonical authority:** §11.4.97.

§11.4.98 — Full-Automation Anti-Bluff: live tests re-runnable without manual intervention

Inherited from constitution/Constitution.md §11.4.98. Every test fully self-driving end-to-end, emits PASS/FAIL/SKIP-with-reason without further human action after startup. Only exception: one-time credential bootstrap OUTSIDE test execution. Re-runnability proof = PASS at `-count=3` consecutive automated invocations. **Canonical authority:** §11.4.98.